

Homework Grading Report

Student:	Student_10
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	15.0	
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Instructor Assessment

You have laid a solid foundation by importing the data and articulating the purpose of each analytical step. However, the majority of the required code is missing, and your reflection answers, while on the right track, lack specific examples from your work. To earn a higher grade, you need to implement the full set of wrangling functions and tie your written insights directly to the results those functions produce.

What You Did Well

Part 1: R Basics and Data Import (Lesson 1) - Task 1.1: Set Working Directory Part 1: R Basics and Data Import (Lesson 1) - Task 1.2: Load Required Packages Part 1: R Basics and Data Import (Lesson 1) - Task 1.3: Import Datasets Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations Part 4: Data

Issues

To strengthen your data cleaning, you need to explicitly identify missing values and outliers using functions like `is.na()` and `boxplot.stats()`. This matters because undetected quality issues can bias revenue calculations. Improve by adding a step such as `sales_data %>% summarise(missing = sum(is.na(Revenue)), outliers = sum(Revenue %in% boxplot.stats(Revenue)$out))` and then handling them (e.g., impute or remove). The solution shows a `cleaned_sales` dataset where 3 outliers were winsorized.

Your transformation steps lack concrete code examples (`select()`, `filter()`, `arrange()`). Without these, the analysis cannot be reproduced. Add pipe-chained commands like `sales_data %>% select(ProductID, Region, Revenue) %>% filter(Revenue > 0) %>% arrange(desc(Revenue))`. The reference solution produces a sorted table of top-selling products.

When reshaping data, you did not demonstrate `pivot_longer()` or `pivot_wider()`. Reshaping is crucial for tidy analysis and for feeding data into modeling functions. Implement `pivot_longer(cols = starts_with("Month"), names_to = "Month", values_to = "Sales")` and then `pivot_wider(names_from = "Metric", values_from = "Value")`. The correct output yields a long table with one row per product-month.

Your join operations are described but not coded. Proper joins ensure that customer, order, and item information align, preventing orphaned records. Use `left_join(customers, orders, by = "CustomerID")` followed by `inner_join(., order_items, by = "OrderID")`. The solution creates an `integrated_sales` dataframe with 350 rows and no missing keys.

String and date handling are missing from the submitted code. Cleaning text (e.g., `str_trim()`) and parsing dates (e.g., `ymd()`) are essential for accurate time-series analysis. Add code such as `orders %>% mutate(Order_Date = ymd(Order_Date), Clean_City = str_trim(City))`. The exemplar shows a Date column correctly converted to Date class and city names standardized.

Reflection & Critical Thinking

In Question 9.1 you note that there were no missing values or outliers, then explain why cleaning would matter. While you correctly identify the importance of completeness and avoiding skew, you do not demonstrate any actual inspection of the data (e.g., using `summary()` or visual checks), which limits the depth of your answer. For Question 9.2 you describe that grouped analysis revealed better-performing regions and categories and that this informs marketing and inventory decisions. This shows good business intuition, but you do not cite any specific metric or example from your analysis (e.g., "Region West had 15% higher revenue than East"), which would strengthen the insight. Your answer to Question 9.3 repeats the same sentences twice and mixes up the purposes of wide vs. long formats. You correctly identify reporting vs. modeling uses, yet you could improve by linking a concrete scenario (e.g., using wide format for a quarterly sales dashboard and long format for a churn prediction model). In Question 9.4 you accurately differentiate `left_join()` and `inner_join()` and give appropriate business contexts. This response is clear and demonstrates understanding of join semantics, though adding a brief code snippet would show practical mastery. Question 9.5 highlights Lesson 8 as the most valuable because it creates KPIs and categories. You explain why KPI creation matters, but you could deepen the answer by naming a specific KPI you built (e.g., "Revenue per Customer") and how it drives strategic decisions.

Business Application

You connect grouped analysis to targeted marketing and inventory allocation, illustrating that you recognize how segment-level insights can guide resource distribution. Your discussion of wide vs. long formats ties the data structure choice to reporting dashboards versus predictive modeling, showing awareness of how format impacts downstream business use cases.

Recommendations

Complete the missing code sections for each lesson, ensuring that every required function (`mutate`, `summarize`, `group_by`, `pivot_*`, `join`, `stringr`, `lubridate`) is demonstrated with working examples. Run a full end-to-end script that produces the executive summary dataframe, then compare your results to the provided solution to verify accuracy. Expand each reflection answer with specific metrics or code snippets from your analysis to demonstrate concrete evidence of your insights.

Technical Summary

- Completion: 16 out of 16 sections (100%). Calculated score: 100%.
- Completed: Part 1: R Basics and Data Import (Lesson 1) - Task 1.1: Set Working Directory, Part 1: R Basics and Data Import (Lesson 1) - Task 1.2: Load Required Packages, Part 1: R Basics and Data Import (Lesson 1) - Task 1.3: Import Datasets, Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values, Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue, Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue, Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations, Part 4: Data Transformation Part 2 (Lesson 4) - Task 4.2: Calculate Overall Summary Statistics, Part 4: Data Transformation Part 2 (Lesson 4) - Task 4.3: Regional Performance Analysis, Part 4: Data Transformation Part 2 (Lesson 4) - Task 4.4: Product Category Analysis, Part 5: Data Reshaping with `tidyr` (Lesson 5) - Task 5.2: Reshape to Wide Format, Part 6: Combining Datasets with Joins (Lesson 6) - Task 6.1: Join Customers and Orders, Part 6: Combining Datasets with Joins (Lesson 6) - Task 6.2: Join Orders and Order Items, Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.1: Clean Text Data, Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.2: Parse Dates and Extract Components, Part 8: Advanced Wrangling & Business Intelligence (Lesson 8) - Task 8.2: Calculate Business KPIs
- Incomplete: None
- Completion: 8/9 sections (89%). Score: 95%
- Variables found: 22/22
- Output accuracy: 95.7% (22/23 checks passed)
